

DINTAKURTHI ACHUTH

Data Scientist | Generative AI & Machine Learning Engineer

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SUMMARY

Results-driven Data Scientist with 3+ years of experience building and deploying end-to-end ML and Generative AI solutions across the full lifecycle — data engineering, modeling, deployment and monitoring — at enterprise scale for Google. Strong in Python, SQL and Java, with deep hands-on work in LLMs, RAG architectures, vector search, fine-tuning (LoRA/QLoRA) and multi-agent systems. Experienced Java developer who can ship AI features inside production-grade full-stack applications. Focused on turning complex models into measurable business impact.

CORE COMPETENCIES

- **Generative AI engineering:** RAG, agentic workflows, fine-tuning (LoRA/QLoRA), evaluation and guardrails for production LLM systems.
- **Applied ML & analytics:** NLP, computer vision, embeddings and similarity search, statistics and experimentation, data storytelling for stakeholders.
- **Engineering:** Java/Spring Boot and Python services, REST APIs, Docker/Kubernetes, CI/CD and cloud deployment on GCP and AWS.

PROFESSIONAL EXPERIENCE

Data Scientist

04/2023 – Present

Virtusa (Deployed at Google)

Hyderabad, India

- **AI in Dashboard:** Built an LLM-powered natural-language query interface that uses SVM-based intent classification to translate plain-English questions into validated SQL, with schema-aware reasoning and interactive visualizations for non-technical stakeholders.
- **BugSuggestion:** Developed a duplicate bug detection system across 20+ Google products using text-embedding-005 (768-dim) and GCP Vector Search; KNN similarity search surfaces top-K duplicates at submission time, automating triage and accelerating resolution.
- Owned the full ML lifecycle on both products: data pipelines and preprocessing, embedding generation, model selection, evaluation, deployment on GCP and post-release monitoring.
- Designed schema-aware prompting and SQL validation guardrails to reduce hallucinated queries and make LLM output safe for non-technical users.
- Collaborated with engineering, product and triage teams to gather requirements, iterate on relevance and ship production features used across 20+ products.
- Recognised with the **Time Award** (2025) for outstanding contributions, innovation and passion in Generative AI initiatives.

KEY PROJECTS

InsightDesk — AI-Powered Support Platform (Java Full-Stack)

2025

Java 21 • Spring Boot 3 • Spring Security (JWT) • Spring Data JPA • PostgreSQL + pgvector • React • Docker • Python/FastAPI model service

- Designed and built a full-stack ticketing and knowledge platform: layered Spring Boot REST backend with JWT role-based auth, JPA/PostgreSQL persistence and a React dashboard.
- Integrated a RAG pipeline (pgvector semantic search + LLM) via a Python microservice to auto-suggest answers and surface similar past tickets; containerised with Docker Compose.
- Applied clean architecture, DTO validation, global exception handling and JUnit/Mockito tests to keep the Java codebase production-ready.

AI-Powered Amazon-Style Product Listing Generator 2025

OpenAI CLIP • FAISS • BM25 • Claude API • Stable Diffusion / ControlNet • React • FastAPI

- Built a multimodal retrieval system with CLIP + FAISS to retrieve similar products from 50K+ records and auto-generate Amazon-style listings via the Claude API.
- Implemented hybrid BM25 + dense retrieval (**Precision@10 +18%**), enhanced visuals with Stable Diffusion/ControlNet, deployed via React + FastAPI; evaluated with MRR and Recall@K.

ResearchPilot — Multi-Agent Research & Analytics Copilot 2025

LangGraph • Claude / OpenAI APIs • RAG (Pinecone + BM25 rerank) • FastAPI • Redis • Docker • GCP

- Built a LangGraph multi-agent system (planner, retriever, SQL-analyst, critic) that answers business questions over documents and warehouse tables, with cited sources and self-verification before responding.
- Added hybrid retrieval with reranking, tool-calling for SQL/Python execution, response caching in Redis and an LLM-as-judge evaluation harness (faithfulness, answer relevance, latency) to track quality across prompt and model changes.
- Exposed the pipeline as an async FastAPI service with streaming responses, request tracing and guardrails (PII masking, prompt-injection checks), containerised for deployment on GCP.

Domain LLM Fine-Tuning with QLoRA 2025

HuggingFace Transformers • PEFT • QLoRA • bitsandbytes • PyTorch • Weights & Biases

- Fine-tuned an open-source LLM with 4-bit QLoRA on a curated instruction dataset, and benchmarked it against the base model and prompted API baselines on task accuracy, cost and latency.
- Built the data cleaning, deduplication and train/eval split pipeline, tracked experiments, and served the adapter locally via Ollama for low-cost inference.

Skin Cancer Classification using Convolutional Neural Networks Research

- Published research applying CNN-based image classification to skin lesion diagnosis, covering data preprocessing, model training and evaluation.

TECHNICAL SKILLS

Programming Languages: Python, Java (Core Java, Spring Boot, JPA/Hibernate, JUnit), Scala, SQL, R, JavaScript

Generative AI: LLMs, SLMs, Prompt Engineering, Fine-tuning (LoRA, QLoRA), RAG, LangChain, LangGraph, AI Agents, Multi-Agent Systems, LLMOps, Ollama

Machine Learning & Deep Learning: Regression, Classification, Clustering, Dimensionality Reduction, CNNs, RNNs, LSTMs, GANs, Transformers, Autoencoders, Computer Vision, Object Detection, Image Segmentation, OCR, NER, Sentiment Analysis, Text Classification, PyTorch, TensorFlow, Scikit-learn, HuggingFace

Data Analysis & Big Data: Pandas, NumPy, SciPy, PySpark, Apache Spark (Scala), EDA, Statistics, Hypothesis Testing, A/B Testing, Feature Engineering, Matplotlib, Seaborn, Plotly, Tableau, Power BI

Databases & Cloud: MySQL, PostgreSQL, Redis, MongoDB, Firebase, Pinecone, ChromaDB, FAISS, Weaviate, pgvector, GCP (Vertex AI, Vector Search, DataNexus), AWS (S3, EC2, IAM)

DevOps & Tools: Docker, Kubernetes, Terraform, CI/CD Pipelines, RESTful APIs, FastAPI, Spring Boot, Git

3D & Creative: 3D Asset Design (Blender), Stable Diffusion / ControlNet pipelines

EDUCATION

B.Tech in Computer Science and Engineering 2019 – 2023

Kalasalingam Academy of Research and Education **CGPA: 8.7**

CERTIFICATIONS & ACHIEVEMENTS

Oracle Certified Java Associate — Oracle 2023

Time Award, Virtusa — outstanding contributions, innovation and passion in Generative AI 2025

Research Publication: “Skin Cancer Classification using Convolution Neural Networks”